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गणितसूत्रम्

# GanitSūtram

## Complete Technical Documentation

Version: v1.0

*"Ancient Mathematics. Modern Platform."*

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Organisation: AITDL | aitdl.com

Vikram Samvat: VS 2082 | Gregorian: 2026-03-08

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# GanitSūtram — Complete Technical Documentation ## Platform Architecture, API Reference & Developer Guide **Version:** v1.0 **Author:** Jawahar R Mallah **Org:** AITDL | aitdl.com **Date:** Vikram Samvat VS 2082 | Gregorian 2026-03-08 **Tagline:** "Ancient Mathematics. Modern Platform."

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# SECTION 1 — PLATFORM OVERVIEW

## 1.1 Vision & Mission

GanitSūtram is a living knowledge ecosystem built on Vedic mathematics, pattern intelligence, and the ancient science of numbers.

**Mission:** Make Vedic and classical mathematics accessible, interactive, and intellectually engaging for students, teachers, researchers, and curious minds worldwide.

## 1.2 The Six Websites

| Site | Domain | Purpose | Status |
|---|---|---|---|
| Portal | ganitsutram.com | Homepage, gate, CMS, search | Live |
| Discoveries | discover.ganitsutram.com | Mathematical discovery browser | Live |
| Learning | learn.ganitsutram.com | Courses, practice, leaderboard | Live |
| Knowledge Map | map.ganitsutram.com | D3 interactive knowledge graph | Live |
| Research Lab | lab.ganitsutram.com | Experimental Vedic tools | Live |
| Solver | solve.ganitsutram.com | Live computation engine | Live |


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## 1.3 Technology Stack

| Layer | Technology |
|---|---|
| Frontend | Vanilla JS, HTML5, CSS3, D3.js (v7) |
| Backend | Node.js 20+, Express.js 4 |
| Database | SQLite (dev) / PostgreSQL 18 (prod) |
| Auth | JWT 15min + Refresh Tokens 30d |
| Search | SQLite FTS5 / PostgreSQL tsvector |
| Realtime | Server-Sent Events (SSE) |
| Email | Nodemailer (SMTP) |
| Deploy | Vercel (frontend) + Railway (backend) |
| i18n | Vanilla JS — EN, HI, SA |
| PWA | Web App Manifest + Service Workers |
| Security | Helmet, express-rate-limit, xss, |
| | IP blacklist, threat detector |
| CDN/WAF | Cloudflare (proxy + DDoS + WAF) |

## 1.4 Complete Build History

| # | Phase | Key Deliverables | Tests |
|---|---|---|---|
| 1 | 6 Websites + UI Core | All sites, design system | — |
| 2 | Core Math Engine | 5 algorithm modules | — |


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| | | | |
|---|---|---|---|
| 3 | Backend API + Database | Express, SQLite, repos | — |
| 4 | Pattern Engine | Kaprekar, Fibonacci | — |
| 5 | School Admin Dashboard | requireRole middleware | — |
| 6 | API Documentation Page | 30+ endpoints docs | — |
| 7 | User Profile + Badges | Profile dashboard | — |
| 8 | Email + Password Reset | Nodemailer, SHA-256 | — |
| 9 | Security Hardening | Helmet, rate limits, XSS | — |
| 10 | Refresh Token Rotation | Family reuse detection | — |
| 11 | Analytics Dashboard | Event telemetry, beacon | — |
| 12 | PWA + Mobile | Manifests, SW, responsive | — |
| 13 | i18n EN/HI/SA | Locale engine, switcher | — |
| 14 | Localised Errors + Numbers | Devanagari formatting | — |
| 15 | End-to-End Test Suite | Custom runner | 96 |
| 16 | Leaderboard + Gamification | 20 badges, points | 104 |
| 17 | CMS | Multilingual editor | 116 |
| 18 | Notification System + SSE | Real-time alerts | 130 |
| 19 | Search + Knowledge Graph | FTS5, BFS pathfinder | 153 |
| 20 | SEO + Launch Prep | OG, JSON-LD, sitemaps | 153 |
| 21 | PostgreSQL + Hybrid Config | Dual-adapter, config.js | 153 |
| 22 | Cloudflare + IP Blacklist | Threat detection | 153 |


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# SECTION 2 — ARCHITECTURE

## 2.1 Root Folder Structure

```
ganitsutram/
├── core/                ← Math algorithms (pure functions)
├── backend/             ← Express API server
├── websites/            ← 6 static sites + ui-core
├── docs/                ← All documentation
└── config/              ← Environment configs
```

## 2.3 System Architecture Diagram

```
┌──────────────────────────────────────────┐
│            User Browser                  │
└──────────────┬───────────────────────────┘
               │ HTTPS
┌──────────────▼───────────────────────────┐
│          Cloudflare Edge                 │
│    DDoS · WAF · CDN · SSL                │
└──────────────┬───────────────────────────┘
               │
      ┌────────┴──────────┐
      │                   │
┌─────▼─────┐         ┌────▼──────┐
│  Vercel   │         │  Railway  │
│ (6 sites) │         │  Backend  │
└─────┬─────┘         └──────┬────┘
      │ fetch()             │
      └──────────┬──────────┘
                 │
┌────────────────▼─────────────────────────┐
│          Express Middleware Stack        │
│   ipBlacklist → helmet → cors →          │
│   sanitise → threatDetector → routes     │
└──────────────┬───────────────────────────┘
               │
```



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```
┌──────────────────▼──────────────────────────┐
│            core/math-engine/                 │
│      Pure functions. No side effects.        │
└──────────────────┬──────────────────────────┘
                   │
┌──────────────────▼──────────────────────────┐
│      SQLite (dev) / PostgreSQL (prod)        │
│           23 tables · WAL mode               │
└──────────────────────────────────────────────┘
```

## 2.4 Layer Rules

1. **Core Immobility**: `core/` must never import from `backend/` or `websites/`.
2. **Pure Math**: Algorithms in `core/math-engine/` must be pure functions.
3. **Frontend Isolation**: Frontend sites only communicate with the Backend API.
4. **Style Sovereignty**: All CSS must reside in `ui-core/`. No ad-hoc styles in individual sites.
5. **Database Abstraction**: Use the repository pattern (`backend/database/`) for all DB interactions.
6. **Security First**: All requests must pass through the security middleware stack.
7. **Stateless API**: Authentication via JWT; no server-side session state (except refresh tokens).
8. **I18n Parity**: All new features must support EN, HI, and SA locales.

## 2.6 Environment Config

The `config/config.js` file handles URL switching between Development and Production environments for all subdomains.

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# SECTION 3 — CORE MATH ENGINE

## 3.1 Overview & Rules

Location: `core/math-engine/`. Every module follows the `validate → compute → return` pattern using CommonJS `module.exports`.



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## 3.2 Module: Digital Root

**Vedic basis:** Beejank, casting out nines
**Sanskrit:** बीजांक
**Functions:** `digitalRoot(n)`, `digitalRootWithSteps(n)`
**Example:** 98 → 9+8=17 → 1+7=8 → result: 8

## 3.3 Module: Squares Ending in 5

**Sutra:** Ekadhikena Pūrvena
**Sanskrit:** एकाधिकेन पूर्वेण
**Functions:** `squaresEndingFive(n)`, `...WithSteps(n)`
**Algorithm:** For n=X5, result = X×(X+1) concat 25
**Example:** $75^2$ → 7×8=56 → 5625

## 3.4 Module: Multiply by 11

**Vedic technique:** Digit insertion method
**Functions:** `multiplyBy11(n)`, `...WithSteps(n)`
**Example:** 57×11 → 5_(5+7)_7 → 627

## 3.5 Module: Nikhilam

**Sutra:** Nikhilam Navatashcaramam Dashatah
**Sanskrit:** निखिलं नवतश्चरमं दशतः
**Functions:** `nikhilam(a, b)`, `...WithSteps(a, b)`
**Example:** 98×97, base 100 → 9506

## 3.6 Module: Ūrdhva-Tiryak

**Sutra:** Ūrdhva-Tiryagbhyām
**Sanskrit:** ऊर्ध्वतिर्यग्भ्याम्
**Functions:** `urdhva(a, b)`, `...WithSteps(a, b)`
**Example:** 23×45 → 1035


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## 3.7 Module: Pattern Engine

| Function | Returns |
|---|---|
| detectDigitalRootCycle | 1-9 repeating cycle analysis |
| detectSquarePattern | [1, 4, 9, 7, 7, 9, 4, 1, 9] cycle |
| detectSumOfDigitsPattern | Linear digit sum analysis |
| detectPalindromePattern | Range density of palindromes |
| detectMultiples11Pattern | Bridging digit sum analysis |
| detectConsecutiveSquareDiff | 2n+1 odd number pattern |
| analyseSequence | AP/GP and property detection |
| kaprekarRoutine | Steps to 6174 or 495 |
| fibonacciDigitalRoots | 24-step DR cycle length |

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# SECTION 4 — BACKEND API

The backend is an Express.js server providing a RESTful interface for all mathematical and platform operations.

## 4.1 Master Endpoint List

- **POST /api/solve**: Central hub for math engine execution.
- **GET /api/concepts**: Knowledge retrieval.
- **GET /api/discoveries**: Pattern library access.
- **POST /api/auth/register-login**: JWT-based identity management.
- **GET /api/leaderboard**: Global gaming ranks.
- **GET /api/notifications**: Real-time SSE channel.


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# SECTION 5 — DATABASE

## 5.1 Dual-Adapter Pattern

Supports transparent switching between **SQLite** (local development) and **PostgreSQL** (production) using a unified `db.js` interface.

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# SECTION 6 — AUTHENTICATION & SECURITY

## 6.1 Multi-Layer Defense

- **IP Blacklist**: Immediate rejection of malicious IPs.
- **Threat Detector**: RT analysis for SQLi, XSS, and Path Traversal.
- **JWT Rotation**: Passive line-hijacking detection via refresh tokens.
- **Rate Limiting**: Intelligent sliding windows per endpoint.

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# SECTION 14 — ATTRIBUTION & LEGAL

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